Digital Preservation of Cultural Heritage using XR Technologies

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0 EXECUTIVE SUMMARY OF THE PROPOSAL

// Rezső innen The project "Digital Preservation of Cultural Heritage using XR Technologies" aims to revolutionize how we interact with and preserve history. By leveraging Extended Reality (XR)—encompassing Virtual, Augmented, and Mixed Reality—this project focuses on creating high-fidelity digital twins of endangered cultural heritage sites. The proposal addresses the urgent need for non-destructive preservation and accessible education. Key outcomes include a standardized framework for digital heritage reconstruction, a cloud-based XR platform for remote exploration, and specialized tools for historical researchers. This summary highlights our interdisciplinary approach combining computer science with digital humanities to ensure that cultural artifacts remain accessible for future generations despite physical decay or environmental threats.

1 TECHNICAL AND SCIENTIFIC DESCRIPTION OF THE PROJECT

1.1 THE PROJECT TOPIC AND ITS PRACTICAL RELEVANCE

Project Topic and Correlation with the Call
The project, "Digital Preservation of Cultural Heritage using XR Technologies," focuses on the high-fidelity digital reconstruction and immersive accessibility of endangered cultural monuments. By integrating 3D laser scanning (LiDAR), aerial photogrammetry, and Extended Reality (XR)—comprising Virtual, Augmented, and Mixed Reality—the proposal creates "digital twins" of historical sites. This aligns with the thematic priorities of the call by fostering technological innovation in the preservation of European cultural identity across Mediterranean, Central-Eastern European, and Atlantic regions.

Practical Relevance
The practical relevance of this research addresses two critical global challenges: the physical degradation of heritage sites caused by environmental factors (e.g., coastal erosion, climate change) and the unsustainable pressure of mass tourism. The project offers a non-destructive preservation method, allowing for the meticulous recording of structures before further decay occurs. Furthermore, it provides a scalable solution for "Virtual-First" tourism, which unburdens fragile UNESCO sites while maintaining economic engagement through digital accessibility.

End-Products and Expected Results
The project will deliver a suite of integrated technological solutions:

- **Standardized Digital Heritage Framework:** A methodological protocol for high-precision 3D documentation in complex outdoor environments.
- **VR-based "Virtual Tour" Engine:** A high-fidelity rendering architecture (Unreal/Unity) capable of handling massive photogrammetry datasets for remote exploration.
- **AR-based "On-Site Time Machine":** A mobile module that overlays digital reconstructions onto real-world ruins in real-time for physical visitors.
- **Open Science Ecosystem:** A cloud-based repository for 3D assets and scientific documentation, ensuring long-term visibility and research integration.

Success Criteria
Success will be measured by the advancement of Technology Readiness Levels (TRL) through validation in real-world ruin scenarios. Key performance indicators include a minimum 85

1.2 PROJECT CONTRIBUTION BEYOND THE STATE OF THE ART

Recent advances in XR technologies and digital twin systems have opened new possibilities in cultural heritage preservation. However, existing solutions are often limited to isolated case studies or indoor environments.

Current limitations include:

- Lack of scalability
- Limited outdoor applicability
- Weak integration between scientific documentation and public accessibility

The project advances the state of the art by:

- Applying XR technologies in complex outdoor archaeological contexts
- Integrating large-scale, multi-region datasets
- Combining scientific documentation with immersive public access

1.3 PROJECT OBJECTIVES AND OUTCOMES

The main objectives of the project are:

- Develop high-quality 3D digital representations of cultural heritage sites
- Enable virtual access through VR/AR technologies
- Reduce physical tourism pressure on vulnerable sites
- Establish standardized methodologies for digital heritage documentation
- Foster interdisciplinary collaboration between technology and humanities
- Support open science through accessible datasets and publications

In detail, the project aims to save lesser-known or rapidly deteriorating monuments using 3D scanning and VR/AR technology, allowing them to be visited without the physical burden of mass tourism. Key targeted achievements include:

- **Technological Validation:** Testing XR platforms in complex, outdoor environments (e.g., ruins, built heritage) to increase Technology Readiness Levels (TRL) and demonstrate real-world impact.
- **Open Science and Visibility:** Generating high-quality datasets, 3D models, and VR assets as part of an open access ecosystem, maximizing visibility, citation, and future funding opportunities.
- **International Network Expansion:** Strengthening EU consortium networks by bringing together technology hubs, scientific publishers, and top-tier heritage protection partners across the Mediterranean, Central-Eastern, and Atlantic European regions.

1.4 ORIGINAL AND INNOVATIVE CONTRIBUTIONS OF THE PROJECT

The project introduces innovative contributions by combining XR technologies with scalable digital heritage workflows.
Key innovations include:

- Use of high-fidelity digital twins as reusable cultural assets
- Integration of XR environments with real-world heritage preservation strategies
- Standardized workflows for outdoor heritage digitization

Further breakthrough areas include:

- **New XR Application Domain:** Shifting advanced 4D reality capture (recording motion, gesture, sound, and physiological data) from industrial and healthcare sectors directly into cultural heritage. This represents a highly innovative use-case that aligns with EU "cultural heritage digitisation" priorities.
- **Validation in Complex Scenarios:** Pushing the technological boundaries by validating scanning technologies on outdoor, deteriorating ruins under real-world conditions, rather than controlled lab environments.
- **Open Access Revenue and Science Models:** Implementing an innovative ecosystem where digital heritage + XR research acts as fuel for an open science platform, increasing accessibility for developers, startups, and museums.

1.5 INTER-, MULTI-, OR TRANS- DISCIPLINARY CHARACTERISTICS

The project adopts an interdisciplinary approach combining computer science, archaeology, cultural heritage studies, and digital humanities. Technological development is closely integrated with heritage expertise, ensuring both scientific accuracy and user accessibility. This approach enables effective collaboration across domains and enhances the societal relevance of the project outcomes.

This interdisciplinary synergy is manifested through the following pillars:

- **Technology and Publishing:** Arts et Métiers (CAPLAB) provides cutting-edge XR tools, while Springer Nature ensures that the resulting digital heritage trend reaches the widest possible academic and public audience through open science and open access mechanisms.
- **Mediterranean Heritage Experts:** Sapienza Università di Roma and the National and Kapodistrian University of Athens bridge classical Greco-Roman archaeology, urban conservation, and climate risk assessments with modern digital documentation methods.
- **European Regional Synergy:** ELTE BTK (Hungary) brings AI-based processing and digital humanities expertise (Danube-AI), while Sorbonne Université (France) coordinates data collection on medieval fortifications and coastal erosion, ensuring a truly pan-European and cross-disciplinary framework.

2 IMPACT AND DISSEMINATION OF THE PROJECT RESULTS

2.1 DISSEMINATION AND EXPLOITATION OF THE PROJECT RESULTS

Dissemination focuses on open science and wide accessibility:

- Open access publications (Springer Nature)
- Public datasets (3D models, scans)
- VR/AR applications for public engagement

Target audiences include:

- Researchers
- Cultural institutions
- Developers and industry stakeholders

2.2 POSSIBLE APPLICATIONS WITH MARKET POTENTIAL

The market potential and exploitation strategies are deeply intertwined with the open science model and XR sector expansion:

- **Open Access Revenue Model:** The partnership with Springer Nature leverages Article Processing Charges (APCs) to ensure free access to publications, 3D models, and VR assets. This attracts not only academics but also software developers and XR startups, leading to higher download rates and increased brand reputation.
- **XR Technology Expansion:** Validating 4D capture in complex cultural heritage scenarios opens a completely new market segment for XR technologies, increasing their societal relevance and commercial viability beyond traditional industrial/health training.
- **Sustainable Tourism Economy:** By addressing the physical destruction caused by overtourism, the project provides a scalable digital alternative. "Experience-replicas" and remote access tools offer tourism boards new digital products that generate engagement while unburdening fragile UNESCO sites.

2.3 ESTIMATED IMPROVEMENTS IN THE QUALITY OF LIFE, WITH RESPECT TO CURRENT PERFORMANCE OF PRODUCTS, TECHNOLOGIES AND/OR SERVICES

The project delivers impact on multiple levels:
Scientific:

- New methodologies in digital heritage
- Interdisciplinary research outputs

Technological:

- Validation of XR technologies
- Advancement of digital twin applications

Societal:

- Reduced tourism pressure
- Increased accessibility to heritage

European:

- Strengthened research networks
- Contribution to EU digital heritage strategies

2.4 PROJECT INTEGRATION IN THE DEVELOPMENT STRATEGY OF PARTNER COMPANIES

The project is strategically aligned with the long-term development goals of the consortium partners, addressing specific institutional and market needs:

- **Springer Nature:** The project integrates directly into their transition from a traditional publisher to a comprehensive **Open Science platform provider**. By treating high-fidelity 3D models and XR assets as citable research outputs, Springer Nature addresses the market need for standardized digital heritage documentation and open access revenue models.
- **Arts et Métiers (CAPLAB / Laval Virtual):** A core strategic goal is to expand their **4D reality capture technologies** from industrial and healthcare sectors into the cultural heritage market. The project addresses their need for technological validation (increasing TRL) in complex, non-controlled outdoor environments, which is essential for their market segment expansion.
- **University Partners (UBB, ELTE, Sapienza, NKUA, Sorbonne):** The project supports their internationalization and **Digital Humanities development strategies**. It addresses the need for interdisciplinary research frameworks that combine classical archaeology with AI-based processing and climate risk assessments.

This synergy ensures that project outcomes are not only scientifically relevant but are also embedded in the sustainable business and research models of all partners beyond the funding cycle.

3 CONSORTIUM DESCRIPTION

3.1 PROJECT DIRECTOR

3.2 CONSORTIUM STRUCTURE

The consortium consists of interdisciplinary partners:
Technology Partner: Arts et Métiers (CAPLAB / Laval Virtual) – XR technologies and 4D data capture
Publishing Partner: Springer Nature – Open access dissemination and scientific publishing
Mediterranean Region:

- Sapienza Università di Roma – archaeology and heritage theory
- National and Kapodistrian University of Athens – outdoor heritage and risk analysis

Central-Eastern Europe:

- Eötvös Loránd University (ELTE) – digital humanities and AI-based heritage processing

Atlantic Region:

- Sorbonne Université – data collection coordination and historical architecture

3.3 PARTNER RESEARCH TEAM LEADERS

Dr. Hősi Rezső Rudolf (Eötvös Loránd University)

- **Age:** 42
- **Career path & Current position:** Senior Researcher at ELTE, Department of Software Technology. PhD in Computer Vision.
- **Professional experience:** 18 years in high-performance computing, 3D rendering architectures, and spatial data processing.
- **Significant achievements:**
  1. Development of the "XR-Render" framework for mobile optimization.
  2. Publication: "Efficient Point-Cloud Processing in Outdoor Environments," IEEE Transactions, 2022.
  3. Leader of the "Digital Heritage Visualization" sub-group at ELTE.
- **Distinctions:** Member of the Hungarian Academy of Sciences (MTA) Public Body.

Prof. Dr. Jean-Luc Soulié (Arts et Métiers)

- **Age:** 52
- **Career path & Current position:** Director of the CAPLAB Platform at Arts et Métiers. Expert in 4D reality capture and XR technologies.
- **Professional experience:** 25+ years in VR/AR development and technology transfer from industrial sectors to cultural heritage.
- **Significant achievements:**
  1. "Laval Virtual" Innovation Award for 4D motion capture in complex outdoor environments.
  2. Prototype: "VEP Engine" (Virtual Europe Platform) for high-fidelity 3D rendering.
  3. Standardized protocol for 4D data capture in fragile UNESCO sites.
- **Distinctions:** Member of the European XR Alliance; French National Order of Merit for Research.

Dr. Helen Zhang (Springer Nature)

- **Age:** 45
- **Career path & Current position:** VP of Open Science Strategy at Springer Nature.
- **Professional experience:** Specialist in scientific publishing, Open Access revenue models, and data-sharing ecosystems for digital assets.
- **Significant achievements:**
  1. Implementation of the "Digital Heritage + XR" Open Access publishing track.
  2. Deliverable: "Final Sustainability and Exploitation Plan" for pan-European research alliances.
  3. White Paper: "Digital Heritage Policy in the EU," presented to EU policymakers in Brussels.

- **Distinctions:** Committee Member of COPE (Committee on Publication Ethics).

Prof. Giulia Rossi (Sapienza Università di Roma)

- **Age:** 49
- **Career path & Current position:** Full Professor of Archaeology at Sapienza. Expert in Mediterranean urban conservation.
- **Professional experience:** Specialist in Greco-Roman monuments and climate risk assessments for UNESCO urban sites.
- **Significant achievements:**
  1. Development of the "Vulnerability and Tourism Impact Index" for historical centers.
  2. Co-author: "Climate Risks in Mediterranean Heritage," Nature Heritage, 2022.
  3. Lead archaeologist for the digital documentation of the Roman Forum ruins.
- **Distinctions:** UNESCO Chair on Urban Heritage Preservation.

Prof. Dr. Dimitri Papadopoulos (National and Kapodistrian University of Athens)

- **Age:** 55
- **Career path & Current position:** Full Professor of Byzantine Archaeology at NKUA.
- **Professional experience:** Specialist in long-term site monitoring and climatic risk analysis for open-air Mediterranean ruins.
- **Significant achievements:**
  1. Lead investigator for the "Climate-Heritage" EU-sensor network for ancient sites.
  2. Publication: "Erosion Patterns in Eastern Mediterranean Marble Structures," Archaeometry Today, 2021.
  3. Developer of the "Aegis-Risk" database for Byzantine fortification monitoring.
- **Distinctions:** Member of the International Council on Monuments and Sites (ICOMOS).

Dr. Claire Lefebvre (Sorbonne Université)

- **Age:** 41
- **Career path & Current position:** Senior Researcher at the Institut d'art et d'archéologie, Sorbonne Université.
- **Professional experience:** Expert in medieval and early modern architectural history, specifically focusing on Atlantic coastal fortifications.
- **Significant achievements:**
  1. Coordination of the "Atlantic Wall" digital documentation project (2018–2022).
  2. Publication: "Coastal Erosion and the Loss of Medieval Heritage," European Journal of Architecture, 2024.
  3. Development of the "Fortress-Data" documentation protocol for military heritage.
- **Distinctions:** Recipient of the Ordre des Arts et des Lettres for cultural heritage research.

3.4 PARTNER TEAM STRUCTURE

P1 – Eötvös Loránd University (ELTE)

- **Team Leader:** Dr. Hősi Rezső Rudolf
- **Key Person:** Oláh Sára (Lead Developer & UX Strategist)
  - **Role:** Oversees the "Danube-AI" integration and the user-centric design of the XR platform.
  - **Experience:** Expert in digital archaeology and AI-based heritage processing.
  - **Achievements:**
    1. UI/UX design for the National Laboratory for Digital Heritage.
    2. Coordination of the "Digital Humanities" project branch.
- **Additional Staff:** 2 AI Engineers (MI-based processing), 1 Archaeologist (Digitization expert), 2 PhD Students (Data collection).

P2 – Arts et Métiers (CAPLAB / Laval Virtual)

- **Team Leader:** Prof. Dr. Jean-Luc Soulié
- **Team Structure:**
  - 1 **4D Capture Specialist** (Knowledge and gesture preservation).
  - 2 **XR Engineers** (VR/AR platform development).
  - 2 **Field Technician** (Outdoor/complex environment scanning).

P3 – Springer Nature

- **Team Leader:** Dr. Helen Zhang
- **Team Structure:**
  - 1 **Open Access Coordinator** (APC and licensing management).
  - 1 **Data Scientist** (3D model and dataset repository).
  - 1 **Brand Manager** (Reputation and social impact).

P4 – Sapienza Università di Roma

- **Team Leader:** Prof. Giulia Rossi
- **Team Structure:**
  - 2 **Classical Archaeologists** (Roman urban history).
  - 1 **Digital Documentation Expert** (Scanning and photogrammetry).
  - 1 **LegalEthical Consultant** (Heritage theory).

P5 – National and Kapodistrian University of Athens (NKUA)

- **Team Leader:** Prof. Dr. Dimitri Papadopoulos
- **Team Structure:**
  - 1 **Environmental Monitoring Expert** (Climate risk analysis).
  - 1 **Byzantine Archaeology Specialist**.

  - 1 **GIS and Mapping Technician** (Outdoor site modeling).

P6 – Sorbonne Université

- **Team Leader:** Dr. Claire Lefebvre
- **Team Structure:**
  - 2 **Architectural Historians** (Medieval fortifications).
  - 1 **Data Collection Coordinator** (Atlantic region lead).
  - 1 **Research Assistant** (Comparative regional data).

3.5 CONSORTIUM COMPLEMENTARITIES AND SYNERGIES BETWEEN PARTNERS

The consortium demonstrates a unique interdisciplinary synergy, combining technical innovation with deep cultural heritage expertise to ensure the project's success. The collaboration is structured around three strategic pillars:

- **Technology and Publishing Synergy:** Arts et Métiers (CAPLAB) provides cutting-edge XR tools and 4D reality capture capabilities (motion, gesture, sound), which are essential for creating high-fidelity digital twins. Springer Nature ensures that these digital assets and research outcomes act as fuel for an innovative Open Science platform, reaching the widest possible academic and public audience through open access mechanisms.
- **Mediterranean Heritage Expertise:** Sapienza Università di Roma and the National and Kapodistrian University of Athens (NKUA) bridge the gap between classical Greco-Roman archaeology and modern digital documentation. Their expertise in urban conservation and climate risk assessment provides the necessary scientific grounding for the technological validation in high-stakes, outdoor environments.
- **Pan-European Regional Synergy:** ELTE BTK (Central-Eastern Europe) contributes specialized AI-based processing and digital humanities expertise through its National Laboratory for Digital Heritage. Sorbonne Université (Atlantic region) coordinates the documentation of medieval fortifications and coastal erosion. Together with the Babeș-Bolyai University (UBB) as the coordinator (CO), this partnership ensures a robust, cross-disciplinary framework that addresses the most pressing heritage challenges across diverse European regions.

This interdisciplinary synergy ensures that the project outcomes are scientifically accurate, technologically advanced, and socially relevant.

4 PROJECT MANAGEMENT

4.1 WORK PLAN, DELIVERABLES AND LOAD BALANCING

This section describes the distribution of tasks and the scientific programme for all partners, broken down into nine consistent work packages (WPs).

Work package list

WP No	Work package title	Work package leader	Start	End
1	Management of the project	Sapienza Università di Roma	1	24
2	Methodological Framework	ELTE BTK	1	6
3	High-Fidelity Data Acquisition	NKUA	4	12
4	XR Platform Development	Arts et Métiers	10	18
5	Overtourism Management	Sorbonne Université	13	21
6	Community Engagement	ELTE BTK	4	24
7	Economic Models	Springer Nature	19	24
8	Testing and Pilot Actions	Arts et Métiers	16	24
9	Sustainability and Diss.	Springer Nature	1	24
TOTAL Person-Months			**252 PM**	

WP1: Project Management and Coordination
WP2: Data collection and fieldwork
WP3: XR development and integration
WP4: XR Platform Development and UX Design

WP no.	4
WP title	XR Platform Development and UX Design
WP leader	Arts et Métiers (CAPLAB / Laval Virtual)
Involved partners	Arts et Métiers, ELTE, UBB, Springer Nature
Start / End month	Month 10 / Month 18
Objectives	**O4.1.** Advanced XR (VR/AR) and 3D integration for heritage. **O4.2.** Preservation of "artisanal intelligence" and 4D capture. **O4.3.** Technological validation (TRL increase) in outdoor environments.
Description of work	**T4.1.** Building the core 3D rendering architecture for high-fidelity photogrammetry datasets (Unreal/Unity). **T4.2.** Iterative usability testing with diverse demographic groups to identify navigation pain points. **T4.3.** Development of an AR module for real-time ruin reconstruction via mobile application .
Deliverables	**D4.1.** VR-based "Virtual Tour" engine (M15) . **D4.2.** UX research and prototype testing report (M15). **D4.3.** AR-based "On-Site Time Machine" module (M18).

WP5: Overtourism Management and Virtual Flow Control

WP no.	5
WP title	Overtourism Management and Virtual Flow Control
WP leader	Sorbonne Université
Involved partners	Sorbonne, Sapienza, NKUA
Start / End month	Month 13 / Month 21
Objectives	**O5.1.** Address mass tourism pressure on vulnerable monuments . **O5.2.** Establish virtual alternatives and divert tourism flows . **O5.3.** Climate-sensitive digitization (coastal erosion monitoring) .
Description of work	**T5.1.** Priority strategic guidelines for digital exploration of vulnerable sites . **T5.2.** Software tools for dynamic physical ticket availability based on sensor data . **T5.3.** Communication strategy to promote sustainable VR/AR alternatives .
Deliverables	**D5.1.** "Virtual-First" tourism model (M18) . **D5.2.** Digital ticketing and capacity monitoring algorithms (M18) . **D5.3.** Awareness campaign on responsible digital tourism (M21) .

WP6: Community Engagement and Social Impact

WP no.	6
WP title	Community Engagement and Social Impact
WP leader	Eötvös Loránd Tudományegyetem (ELTE BTK)
Involved partners	ELTE, Local communities
Start / End month	Month 4 / Month 24
Objectives	**O6.1.** Foster societal connections through digital assets . **O6.2.** Involve local populations and integrate legends into narratives . **O6.3.** Measure social impact of digital heritage interventions .
Description of work	**T6.1.** Interactive sessions with residents and historians on regional digitization . **T6.2.** Enriching digital twins with authentic, crowdsourced oral histories . **T6.3.** Systematic measurement of influence on local economies and conservation attitudes .
Deliverables	**D6.1.** Community workshop series (M12) . **D6.2.** Co-creation digital archive (M12) . **D6.3.** Social Impact Assessment (SIA) report (M24) .

WP7: Economic Models and Revenue Generation

WP no.	7
WP title	Economic Models and Revenue Generation
WP leader	Springer Nature
Involved partners	Springer Nature, Local communities
Start / End month	Month 19 / Month 24
Objectives	**O7.1.** Establish financial framework for self-sufficiency . **O7.2.** Transform digital twins into economic assets for communities .
Description of work	**T7.1.** Tiered access system (Basic/Premium/Seasonal) for the VR platform . **T7.2.** Standardized licensing packages for schools and museums . **T7.3.** Digital bridge for purchasing physical local artisanal products .
Deliverables	**D7.1.** Comprehensive Business Sustainability Plan (M24) . **D7.2.** Standardized Licensing Agreement Template (M24) .

WP8: Testing, Validation and Pilot Actions

WP no.	8
WP title	Testing, Validation and Pilot Actions
WP leader	Arts et Métiers (CAPLAB / Laval Virtual)
Involved partners	Arts et Métiers, ELTE, Sapienza, NKUA, Sorbonne
Start / End month	Month 16 / Month 24
Objectives	**O8.1.** Technical and social validation of prototypes . **O8.2.** Measure resilience to environmental factors and impact on preservation .
Description of work	**T8.1.** Multi-regional pilot events (Launch Days) across four European countries . **T8.2.** Evaluation of rendering performance across diverse VR/AR hardware . **T8.3.** UX research including eye-tracking and qualitative focus groups .
Deliverables	**D8.1.** Pilot Evaluation Report (M21) . **D8.2.** Social Impact and UX Assessment Report (M24) .

WP9: Sustainability, Dissemination and Communication

WP no.	9
WP title	Sustainability, Dissemination and Communication
WP leader	Springer Nature
Involved partners	Springer Nature, All Partners
Start / End month	Month 1 / Month 24
Objectives	**O9.1.** Post-project life cycle sustainability (CHARM-EU model) . **O9.2.** Permanent legal identity and IP management . **O9.3.** Scientific dissemination of results .
Description of work	**T9.1.** Peer-reviewed publications and conference presentations (EuroVR, Digital Heritage) . **T9.2.** Flagship closing exhibition in Brussels for EU policymakers . **T9.3.** Creation of a separate legal identity (AISBL) for long-term repository management .
Deliverables	**D9.1.** Final Sustainability and Exploitation Plan (M24) . **D9.2.** Communication Toolkit and branding package (M12) .

List of Deliverables

Del. no.	Deliverable name	WP	Leader	Nat.*	Diss.**	Date
1.1	Admin/Financial Report	1	Sapienza	RE	CO	24
2.1	Vulnerability Index	2	ELTE	IS	PU	3
3.1	LiDAR/Photogrammetry Dataset	3	NKUA	DA	PU	6
4.1	VR "Virtual Tour" Engine	4	AetM	IT	PU	15
5.1	Tourism Model Strategy	5	Sorbonne	IS	PU	18
6.3	SIA Report	6	ELTE	RE	PU	24
9.1	Scientific Publications	9	Springer	IS	PU	24

**Nature: RE=Report, IS=Intellectual Service, DA=Dataset, IT=IT tool. **Diss: PU=Public, CO=Confidential.*

4.2 COORDINATION AND TASK SCHEDULE

The project timeline spans 24 months and is structured through a phased implementation of Work Packages.

A detailed Gantt chart defines task dependencies and scheduling across all WPs.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP 1: Management of the project	1.1	Administrative and financial management	Sapienza Università di Roma								
	1.2	Development of a Quality Assurance and Risk Management Plan									
	1.3	Data Management Plan (DMP) based on GDPR and Open Science principles									
WP2: Methodological Framework and Site Selection	2.1	Development of a vulnerability and tourism impact index.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	2.2	Multi-stakeholder analysis (involving local population, local governments, academia)									
	2.3	Finalization of site selection matrix (aspects: degree of deterioration, limited accessibility).									
WP3: High-Fidelity Data Acquisition and 3D Scanning	3.1	Laser scanning (LiDAR) and aerial photogrammetry (drone-based data collection).	Nemzeti Kapodisztriasz Athéni Egyetem (NKUA)								
	3.2	Texture optimization and material-specific digital preservation.									
	3.3	Metadata enrichment (integration of archaeological and art historical data).									
WP4: XR Platform Development and UX Design	4.1	Development of a VR-based "Virtual Tour" engine (Unreal Engine/Unity).	Arts et Métiers (CAPLAB / Laval Virtual)								
	4.2	User Experience (UX) research and prototype testing with focus groups.									
	4.3	Development of an AR-based "On-Site Time Machine" module (ruin reconstruction via mobile application).									

Figure 1: Project schedule (Gantt chart) of WP1, WP2, WP3 and WP4 for the 24-month period.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP5: Overtourism Management and Virtual Flow Control	5.1	Development of a "Virtual-First" tourism model (remote access vs. on-site restrictions).	Sorbonne Université								
	5.2	Design of digital ticketing system and "capacity monitoring" algorithms.									
	5.3	Awareness campaign about the "responsible digital tourist" concept.									
WP6: Community Engagement and Social Impact	6.1	Community workshops on the relationship between local identity and digital representation.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	6.2	Co-creation processes: integrating local legends into the XR narrative.									
	6.3	Preparation of a social impact assessment (SIA).									
WP7: Economic Models and Revenue Generation	7.1	Digital ticketing and subscription models (B2C).	Springer Nature								
	7.2	Development of a licensing framework for educational institutions and museums (B2B).									
	7.3	Integrating local artisanal products and services into the virtual marketplace.									
WP8: Testing, Validation and Pilot Actions	8.1	Organizing pilot events in 4 countries (on-site and remote testing).	Arts et Métiers (CAPLAB / Laval Virtual)								
	8.2	Technical performance evaluation (loading speed, rendering accuracy).									
	8.3	Collect user feedback using quantitative and qualitative methods.									
WP9: Sustainability, Dissemination and Communication	9.1	Scientific publications and conference participation (Open Access).	Springer Nature								
	9.2	Organization of a closing conference and "Virtual Heritage" exhibition.									
	9.3	Development of an Exploitation Plan to adapt the technology to other regions.									

Figure 2: Project schedule (Gantt chart) of WP5, WP6, WP7, WP8 and WP9 for the 24-month period.

4.3 AVAILABLE RESEARCH INFRASTRUCTURE, AND ITS UPGRADE/DEVELOP

The project relies on a comprehensive network of specialized infrastructures provided by the consortium:

- **CAPLAB Platform (Arts et Métiers / Laval Virtual):** A European hub for XR, digital twins, and 3D technologies, providing robust infrastructure for 4D reality capture and technology transfer.
- **Digital Open Science Ecosystem (Springer Nature):** The platform infrastructure for hosting and distributing open access articles, datasets, 3D models, and VR assets.

- **Mediterranean Field Sites (Sapienza & NKUA):** Access to highly vulnerable urban UNESCO sites in Rome and weather-exposed open-air ruins in Greece, serving as the physical infrastructure for technological testing and validation.
- **Digital Humanities Network (ELTE BTK):** The National Laboratory for Digital Heritage and the Danube-AI initiative in Hungary, supplying the computational and methodological infrastructure for AI-driven heritage processing.
- **Data Collection Network (Sorbonne Université):** Established infrastructure for coordinating architectural and historical documentation, particularly concerning fortifications and coastal heritage.

4.4 MANPOWER ALLOCATION

List of research team members		**Position in the project**	**PM****	**Cost (RON)**
Coordinator (UBB)	Dr. Krémer Aliz	Project Director	18	120,000
	[Vacant]	Senior Software Engineer	12	80,000
	[Vacant]	Financial/Admin Manager	12	60,000
	[Vacant]	PhD Researchers (2)	24	90,000
Partner 1 (ELTE)	Dr. Hősi Rezső Rudolf	P1 Responsible	12	70,000
	Oláh Sára	Key Researcher (UX)	10	60,000
	[Vacant]	AI Engineers (2)	16	90,000
	[Vacant]	PhD Researchers (2)	12	40,000
Partner 2 (AetM)	Prof. Dr. Jean-Luc Soulié	P2 Responsible	10	50,000
	[Vacant]	XR Engineers (2)	20	40,000*
	[Vacant]	4D Capture Specialist	12	30,000*
Partner 3 (Springer)	Dr. Helen Zhang	P3 Responsible	6	25,000
	[Vacant]	Open Access Coordinator	12	15,000*
Partner 4 (Sapienza)	Prof. Giulia Rossi	P4 Responsible	12	40,000
	[Vacant]	Archaeologists (2)	18	20,000*
Partner 5 (NKUA)	Prof. Dr. Dimitri Papadopoulos	P5 Responsible	12	40,000
Partner 6 (Sorbonne)	Dr. Claire Lefebvre	P6 Responsible	12	40,000
TOTAL			**252**	**800,000**

*Own budget contribution (Partner share).
*To be indicated with respect to the total project duration (24 months).
Indicate also the non-permanent staff (vacant positions) to hire – doctoral and post-doctoral researchers, interns, etc.

4.5 PROJECT BUDGET AND PARTNERS SHARE

The budget is allocated across major categories:

- Personnel costs
- Equipment and technology (XR, scanning tools)

- Travel and fieldwork
- Dissemination and publication

Detailed financial planning ensures efficient resource allocation and alignment with project objectives.

5 REFERENCES